
2019-Ph-H1-Q17

Section: Our Dynamic Universe

Topic: Gravitation

Question:

The gravitational field strength on the surface of a planet is 20 N/kg. What is the acceleration due to gravity on the planet?

Answer:

Gravitational field strength and gravitational acceleration are numerically equal:

$$g = 20 \text{ N/kg} = 20 \text{ m/s}^2$$

Correct Option: D (20 m/s²)

Guidance for Students:

Field strength g is the acceleration experienced by a 1 kg mass — same as acceleration due to gravity.

Revision Tips:

- Units of field strength: $\text{N/kg} = \text{m/s}^2$
 - Don't confuse with gravitational force (in newtons)
-